

Day 6 — Why It Works — The Algebra

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive $(b-a)(b-b) = b(b-a-b) + ab$.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Derive the formula.** Let B = the base (10, 100, 1000). Factor 1 = $B - a$, Factor 2 = $B - b$.
$$(B - a)(B - b) = B^2 - Ba - Bb + ab = B(B - a - b) + ab = B \cdot (\text{left part}) + (\text{right part})$$

So the left part is $B - a - b = (B - a) - b = (B - b) - a = \text{either factor minus the OTHER’s deficit}$. The right part is just ab .
2. **Why “cross-subtract”?** $B - a - b = \text{first factor} - (B - \text{second factor}) = \text{first factor} - \text{second’s deficit}$. The cross-subtraction is just rearranging the algebra into a mental-math procedure.
3. **Above-base proof.** $(B + a)(B + b) = B^2 + Ba + Bb + ab = B(B + a + b) + ab$. Cross-add, product of excesses.
4. **Mixed case proof.** $(B + a)(B - b) = B^2 - Bb + Ba - ab = B(B + a - b) - ab$. The “ $-ab$ ” in the right slot is what causes the sign issue.
5. **Connecting the dots.** This is just basic high-school algebra — no mystery. Tirthaji’s sūtra system gives the procedure a memorable name; the algebra gives it the proof.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).